FORM 2

THE PATENT ACT, 1970

(39 of 1970)

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The Patent Rules, 2003

COMPLETE SPECIFICATION

(See section 10 and rule 13)

1. TITLE OF THE INVENTION:

A System and Method for Tokenless Laundry Bag Identity Generation, Localization, Verification and Recovery

2. APPLICANT:

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3. PREAMBLE TO THE DESCRIPTION:

The following specification particularly describes the invention and the manner in which it is to be performed:

4. DESCRIPTION:

Field of the invention:

[0001] The present disclosure generally relates to the technical field of artificial intelligence (AI), computer vision, and smart laundry management systems, in specific, relates to a system and method for tokenless laundry bag identity generation, autonomous rack localization, collection verification, and misplaced laundry bag recovery in large-scale laundry environments.

Background of the invention:

[0002] Laundry management in universities, hostels, hospitals, hotels, institutional facilities, and commercial laundry environments involves handling, processing, storing, locating, and returning a large number of laundry bags associated with different users. Conventional laundry management systems commonly employ physical identifiers, such as radio-frequency identification (RFID) tags, QR codes, barcodes, or physical tokens, for associating laundry bags with respective users and tracking the laundry bags during processing and collection. Such systems generally require continued availability and readability of the respective physical identifiers and may also involve predetermined sorting or manual rack-management procedures for locating processed laundry bags.

[0003] Various prior art references disclose systems for managing and tracking laundry items using machine-readable identifiers. For instance, IN202611028832A, titled “Intelligent QR Code Enabled Automated Laundry Management and Tracking System,” discloses an institutional laundry-management arrangement in which a unique bag identification value is associated with a registered user, garment-count and laundry-type information is received through a digital submission interface, and a unique QR code is generated and mapped to the respective bag information. The laundry-processing status is updated during the laundry cycle and made available to the user through a digital portal. Although such an arrangement facilitates digital tracking of laundry operations, identification and tracking remain dependent on a machine-readable QR code associated with the laundry bag. Consequently, damage, detachment, obscuring, or unavailability of the machine-readable identifier may impair reliable association of the physical laundry bag with the corresponding user information.

[0004] Another prior art, IN202541067150A, titled “Integrated RFID and Vision-Based Smart Laundry Management System for Automated Garment Tracking, Classification, and Billing,” discloses a laundry-management system employing RFID tracking, machine-vision techniques, connected devices, and cloud-based data processing. A camera arrangement associated with a processing unit and machine-learning models is utilized for garment recognition and counting, while RFID tagging enables traceability and status communication through interfaces such as a mobile application or web portal. Although the system integrates vision-based recognition with digital laundry management, the tracking architecture remains dependent on RFID-based identification for maintaining traceability and does not eliminate the technical dependency associated with attached or readable physical identifiers.

[0005] Systems directed to tagless identification have also been proposed outside the laundry-management field. For example, US20180111699A1, titled “Tagless Baggage Tracking System and Method,” discloses generation of a searchable baggage record using a digital image together with information such as weight, colour, size, dimensions, shape, brand, and baggage type. The stored image and associated weight information can be searched for identifying an owner of tagless baggage or locating misdirected or lost baggage. While this approach demonstrates identification of baggage using visual and physical characteristics without exclusive reliance on an attached tag, it is directed to baggage handling and does not address the technical requirements associated with continuously identifying and determining the storage position of numerous visually similar laundry bags distributed across multi-level storage racks in a high-volume laundry environment.

[0006] Another prior art, US20220051177A1, titled “System and Method for Identifying Misplaced Products in a Shelf Management System,” describes machine-vision-based identification of misplaced products using images acquired from shelf environments. Extracted image features are compared with stored product-library features to obtain a best-fit identification, and fixed-position cameras can be employed for monitoring shelf areas. Related non-patent literature concerning product recognition on retail racks additionally describes object-localization and image-encoding techniques using colour, texture, and confidence-based product-identification approaches. Such systems are primarily directed to recognizing known product classes or stock arrangements in retail environments and do not address the technical difficulties arising when individual deformable laundry bags associated

with different users are visually similar, are positioned without a predefined storage sequence, and undergo repeated handling between processing, storage, and collection stages.

[0007] Conventional identification and storage-management architectures therefore present several technical limitations when applied to large-scale laundry environments. Identifier-dependent systems can lose tracking continuity when an RFID tag, QR code, barcode, or token becomes damaged, detached, obscured, or unreadable. Systems based primarily on predetermined storage organization require continued conformity between the recorded location and the actual physical position of a laundry bag. Further, conventional image-recognition arrangements developed for retail products or baggage do not necessarily provide reliable continuity between user association, storage-position determination, collection-stage validation, and subsequent handling of a laundry bag whose position changes after an initial location has been recorded.

[0008] In practical laundry facilities, processed laundry bags can be handled by multiple workers, placed at different positions within racks, shifted from one shelf or slot to another, or retrieved from a location different from an earlier recorded position. Such environments can employ centralized or distributed cameras, sensing devices, processing units, databases, user interfaces, and independently deployed functional modules. The resulting variation in storage configuration, bag appearance, bag orientation, occlusion, position, and handling condition create a technical challenge in maintaining reliable correspondence between a particular physical laundry bag, the associated user information, and the current physical storage location throughout the laundry cycle.

[0009] Further, incorrect retrieval of a laundry bag or absence of an expected laundry bag from a recorded position can require manual inspection of multiple storage racks and repeated comparison of similar-looking bags. In high-volume facilities, such manual intervention can interrupt the continuity of the tracking process and make it difficult to determine whether the retrieved physical bag corresponds to the previously registered bag information. Existing architectures therefore leave an unresolved technical problem concerning reliable identification, localization, validation, and subsequent tracing of laundry bags under dynamically changing and non-predefined storage conditions.

[0010] Therefore, there is a need for a system and method for tokenless laundry bag identity generation, autonomous rack localization, collection verification, and misplaced laundry bag recovery in large-scale laundry environments. There is also a need for a system capable of maintaining reliable association between a user and a corresponding physical laundry bag without dependence on RFID tags, QR codes, barcodes, or physical tokens, particularly when laundry bags are randomly arranged or relocated within storage racks. There is also a need for a system capable of reliably verifying a retrieved laundry bag and identifying a misplaced or incorrectly positioned laundry bag while reducing manual rack inspection, sorting, and searching operations.

Objectives of the invention:

[0011] The primary objective of the present invention is to provide a system and method for tokenless laundry bag identity generation, autonomous rack localization, collection verification, and misplaced laundry bag recovery in large-scale laundry environments.

[0012] Another objective of the present invention is to provide a system that generates a Laundry Identity Profile for each laundry bag based on visual and physical characteristics of the laundry bag and associates the Laundry Identity Profile with corresponding user information.

[0013] Another objective of the present invention is to provide a system that identifies and tracks laundry bags without dependence on radio-frequency identification (RFID) tags, QR codes, barcodes, and physical tokens.

[0014] Another objective of the present invention is to provide a system configured to identify laundry bags positioned on storage racks without a predefined sorting sequence by using computer vision-based detection, feature extraction, and similarity matching.

[0015] Another objective of the present invention is to provide a system that autonomously determines a storage location of each identified laundry bag in terms of rack number, shelf number, and slot position.

[0016] Another objective of the present invention is to provide a system that is configured to verify a retrieved laundry bag during collection by comparing visual and physical

characteristics of the retrieved laundry bag with a corresponding stored Laundry Identity Profile.

[0017] Another objective of the present invention is to provide a system that determines a confidence score corresponding to a retrieved laundry bag based on feature similarity and weight similarity for validating collection of the laundry bag.

[0018] Another objective of the present invention is to provide a system that detects misplaced, incorrectly positioned, and incorrectly retrieved laundry bags based on discrepancies between expected laundry bag information, stored location information, and collection verification results.

[0019] Another objective of the present invention is to provide a system that performs rack-wide searching and re-identification of a misplaced laundry bag based on comparison of candidate laundry bags with a corresponding stored Laundry Identity Profile.

[0020] Another objective of the present invention is to provide a system that determines an updated storage location of a re-identified laundry bag and synchronize the updated storage location with a centralized database and a mobile application.

[0021] Yet another objective of the present invention is to provide a system that provides laundry submission status, laundry processing status, collection availability, and storage location information to a user through the mobile application.

[0022] Further objective of the present invention is to provide a system that reduces manual sorting, rack inspection, and searching operations while supporting reliable handling of a large number of laundry bags in university, institutional, and commercial laundry environments.

Summary of the invention:

[0023] The present disclosure proposes a system and method for tokenless laundry bag identity generation, localization, verification and recovery. The following presents a simplified summary in order to provide a basic understanding of some aspects of the claimed subject matter. This summary is not an extensive overview. It is not intended to identify key/critical elements or to delineate the scope of the claimed subject matter. Its sole purpose is to

present some concepts in a simplified form as a prelude to the more detailed description that is presented later.

[0024] In order to overcome the above deficiencies of the prior art, the present disclosure is to solve the technical problem to provide a system and method for tokenless laundry bag identity generation, autonomous rack localization, collection verification, and misplaced laundry bag recovery in large-scale laundry environments.

[0025] According to one aspect, the invention provides a system for tokenless laundry bag identity generation, localization, verification and recovery. In one embodiment herein, the system comprises a computing device having at least one processor and a memory for storing one or more instructions executable by the at least one processor. In one embodiment herein, the computing device is communicatively coupled to a server via a network. In one embodiment herein, the computing device is communicatively coupled to one or more cameras and at least one weight sensor arranged to acquire visual and physical characteristics of a laundry bag.

[0026] In one embodiment herein, the at least one processor is configured to execute plurality of modules for identifying, localizing, verifying, and recovering the laundry bag based on the acquired visual and physical characteristics of the laundry bag. In one embodiment herein, the plurality of modules comprises a registration module, a feature acquisition module, an artificial intelligence (AI) identity generation module, a centralized database module, an AI-based rack localization module, a user interaction module, a collection verification module, and a recovery module.

[0027] In one embodiment herein, the registration module is configured to associate the laundry bag with a corresponding user identity based on at least one of a registration number, an identity card, and authentication through a user device. In one embodiment herein, the feature acquisition module is configured to receive the visual and physical characteristics of the laundry bag acquired through the one or more cameras and the at least one weight sensor. In one embodiment herein, the visual and physical characteristics comprise image appearance, colour distribution, shape, dimensions, texture characteristics, and weight.

[0028] In one embodiment herein, the artificial intelligence (AI) identity generation module is configured to extract features from the visual and physical characteristics, convert the extracted features into numerical feature vectors, and generate a Laundry Identity Profile associated with the corresponding user identity. In one embodiment herein, the artificial intelligence (AI) identity generation module is configured to assign respective weighting factors to the extracted features and combine similarity values of the extracted features according to the respective weighting factors for determining the feature similarity score.

[0029] In one embodiment herein, the centralized database module is configured to store the Laundry Identity Profile in association with the corresponding user identity, laundry-processing status, rack-location information, collection history, and verification and recovery records. In one embodiment herein, the centralized database module is configured to synchronize the Laundry Identity Profile, the laundry-processing status, the rack-location information, the collection history, and the verification and recovery records with the AI-based rack localization module and the user interaction module.

[0030] In one embodiment herein, the AI-based rack localization module is operatively connected to the centralized database module. The AI-based rack localization module is configured to receive rack image data acquired by one or more overhead cameras arranged to monitor storage racks containing plurality of laundry bags positioned without a predefined sorting sequence. In one embodiment herein, the AI-based rack localization module is configured to perform multi-object laundry bag detection on the rack image data, extract visible characteristics of detected laundry bags, compare the visible characteristics with stored Laundry Identity Profiles using similarity matching, identify the laundry bag based on the similarity matching, and determine a rack number, a shelf number, and a slot position corresponding to the identified laundry bag.

[0031] In one embodiment herein, the one or more overhead cameras are configured to continuously acquire rack image data from respective sections of the storage racks. In one embodiment herein, the AI-based rack localization module is configured to identify the laundry bag based on the highest similarity match between the visible characteristics of the detected laundry bags and the stored Laundry Identity Profiles.

[0032] In one embodiment herein, the user interaction module is configured to receive the rack number, the shelf number, and the slot position determined by the AI-based rack localization module and provide laundry-processing status, collection availability status, and location information associated with the laundry bag to the corresponding user through the user device. In one embodiment herein, the user interaction module is configured to provide the corresponding user with laundry submission confirmation, laundry-processing status, laundry-completion notification, the rack number, the shelf number, the slot position, and collection availability status associated with the laundry bag.

[0033] In one embodiment herein, the collection verification module is configured to compare current visual and physical characteristics of a retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity. In one embodiment herein, the collection verification module is configured to determine a feature similarity score and a weight similarity score, generate a confidence score based on the feature similarity score and the weight similarity score, and authorize collection of the retrieved laundry bag when the confidence score satisfies a predefined threshold.

[0034] In one embodiment herein, the collection verification module is configured to determine the weight similarity score based on a difference between an originally recorded weight of the laundry bag and a current measured weight of the retrieved laundry bag relative to the originally recorded weight. In one embodiment herein, the collection verification module is configured to generate the confidence score by combining the feature similarity score and the weight similarity score according to respective weighting constants. In one embodiment herein, the collection verification module is configured to update a collection status associated with the laundry bag through the centralized database module upon authorization of collection based on satisfaction of the predefined threshold.

[0035] In one embodiment herein, the recovery module is configured to initiate rack-wide scanning of the storage racks in response to detection of at least one condition selected from absence of the laundry bag from a determined rack location and failure of the confidence score associated with the retrieved laundry bag to satisfy the predefined threshold. In one embodiment herein, the recovery module is configured to compare candidate laundry bags detected during the rack-wide scanning with the stored Laundry Identity Profile, determine

respective confidence scores associated with the candidate laundry bags, select a candidate laundry bag having the highest confidence score, and determine a corrected rack number, the shelf number, and the slot position of the selected candidate laundry bag.

[0036] In one embodiment herein, the recovery module is configured to update the corrected rack-location information through the centralized database module and provide the corrected rack-location information to the corresponding user through the user interaction module for subsequent collection of the laundry bag. In one embodiment herein, the recovery module is configured to initiate recovery automatically upon detection of at least one condition selected from absence of the laundry bag from the determined rack location and a mismatch between characteristics of the retrieved laundry bag and the stored Laundry Identity Profile. In one embodiment herein, the recovery module is further configured to initiate recovery in response to a missing-bag report received through the user device.

[0037] In one embodiment herein, identification, localization, verification, and recovery of the laundry bag are performed independently of an RFID tag, a barcode label, a QR code, and other physical identifiers associated with the laundry bag. The artificial intelligence (AI) identity generation module, the centralized database module, the AI-based rack localization module, the user interaction module, the collection verification module, and the recovery module operate in combination to provide tokenless identification, localization, verification, and recovery of the laundry bag based on the visual and physical characteristics associated with the laundry bag.

[0038] According to another aspect, the invention provides a method for tokenless laundry bag identity generation, localization, verification, and recovery using the system. At one step, the registration module associates the laundry bag with the corresponding user identity based on identification information received through the user device. At a subsequent step, the feature acquisition module acquires visual and physical characteristics of the laundry bag through one or more cameras and at least one weight sensor. At a next step, the artificial intelligence (AI) identity generation module generates the Laundry Identity Profile from numerical feature vectors derived from the visual and physical characteristics of the laundry bag, and the centralized database module stores the Laundry Identity Profile.

[0039] At a subsequent step, the AI-based rack localization module receives rack image data acquired by one or more overhead cameras arranged to monitor storage racks containing plurality of laundry bags positioned without the predefined sorting sequence. At a next step, the AI-based rack localization module detects laundry bags from the rack image data, compares visible characteristics of the detected laundry bags with stored Laundry Identity Profiles using similarity matching, and identifies the laundry bag. At a subsequent step, the AI-based rack localization module determines the rack number, the shelf number, and the slot position corresponding to the identified laundry bag, and the user interaction module provides location information associated with the identified laundry bag to the corresponding user through the user device.

[0040] At a next step, the collection verification module verifies the retrieved laundry bag based on comparison of current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile, generates the confidence score based on the feature similarity score and the weight similarity score, and authorizes collection of the retrieved laundry bag when the confidence score satisfies the predefined threshold.

[0041] At a subsequent step, the recovery module initiates rack-wide scanning of the storage racks upon detection of at least one condition selected from absence of the laundry bag from the determined rack location and failure of the confidence score to satisfy the predefined threshold, compares candidate laundry bags detected during the rack-wide scanning with the stored Laundry Identity Profile, and determines respective confidence scores associated with the candidate laundry bags. At a further step, the recovery module selects the candidate laundry bag having the highest confidence score, determines the corrected rack number, the shelf number, and the slot position of the selected candidate laundry bag, updates corrected rack-location information through the centralized database module, and provides the corrected rack-location information to the corresponding user through the user interaction module.

[0042] Further, objects and advantages of the present invention will be apparent from a study of the following portion of the specification, the claims, and the attached drawings.

Detailed description of drawings:

[0043] The accompanying drawings, which are incorporated in and constitute a part of the specification, illustrate an embodiment of the invention, and, together with the description, explain the principles of the invention.

[0044] FIG. 1 illustrates a block diagram of a system for tokenless laundry bag identity generation, localization, verification, and recovery, in accordance to an exemplary embodiment of the invention.

[0045] FIG. 2 illustrates a functional block diagram of the system for tokenless laundry bag identity generation, localization, verification, and recovery, in accordance to an exemplary embodiment of the invention.

[0046] FIG. 3 illustrates a process flow diagram of the system for tokenless laundry bag identity generation, localization, verification, and recovery, in accordance to an exemplary embodiment of the invention.

[0047] FIG. 4 illustrates a state transition diagram of the system for tokenless laundry bag identity generation, localization, verification, and recovery, in accordance to an exemplary embodiment of the invention.

[0048] FIGs. 5A and 5B illustrate a flowchart of a method for tokenless laundry bag identity generation, localization, verification, and recovery using the system, in accordance to an exemplary embodiment of the invention.

Detailed invention disclosure:

[0049] Various embodiments of the present invention will be described in reference to the accompanying drawings. Wherever possible, same or similar reference numerals are used in the drawings and the description to refer to the same or like parts or steps.

[0050] The present disclosure has been made with a view towards solving the problem with the prior art described above, and it is an object of the present invention to provide a system 100 and method for tokenless laundry bag identity generation, autonomous rack localization, collection verification, and misplaced laundry bag recovery in large-scale laundry environments.

[0051] According to an exemplary embodiment of the invention, FIG. 1 refers to a block diagram of a system 100 for tokenless laundry bag identity generation, localization, verification, and recovery. In one embodiment herein, the system 100 identifies and tracks laundry bags without dependence on radio-frequency identification (RFID) tags, QR codes, barcodes, and physical tokens. The system 100 detects misplaced, incorrectly positioned, and incorrectly retrieved laundry bags based on discrepancies between expected laundry bag information, stored location information, and collection verification results. The system 100 reduces manual sorting, rack inspection, and searching operations while supporting reliable handling of a large number of laundry bags in university, institutional, and commercial laundry environments.

[0052] In one embodiment herein, the system 100 incorporates a computing device 102 having at least one processor 104 and a memory 106 for storing one or more instructions executable by the at least one processor 104. The computing device 102 is communicatively coupled to a server 108 through a network 110. Data exchange between the computing device 102 and the server 108 occurs through the network 110 during operation of the system 100. In one embodiment herein, the computing device 102 is implemented as at least one of a smartphone, a mobile phone, a computer, a tablet personal computer, a laptop computer, a wearable device, a personal digital assistant, and an Internet of Things device configured to execute the operations associated with the system 100.

[0053] In one embodiment herein, the network 110 supports wired communication or wireless communication between the computing device 102 and the server 108. The wired communication is implemented through at least one of a Local Area Network (LAN), a Wide Area Network (WAN), a Controller Area Network (CAN), and Ethernet. The wireless communication is implemented through at least one of a cellular network, wireless Local Area Network (Wi-Fi), Bluetooth, Bluetooth Low Energy, Zigbee, Wi-Fi Direct, Ultra-Wideband (UWB), Infrared Data Association (IrDA), and Near Field Communication (NFC).

[0054] In one embodiment herein, the at least one processor 104 is implemented through at least one of a Central Processing Unit (CPU), an Application-Specific Integrated Circuit (ASIC), an Application-Specific Instruction-Set Processor (ASIP), a Graphics Processing Unit (GPU), a Digital Signal Processor (DSP), a Field Programmable Gate Array (FPGA), a Programmable

Logic Device (PLD), a controller, a microcontroller unit, a microprocessor, and an ARM-based processor. The at least one processor 104 incorporates one or more processing cores configured to execute the instructions stored in the memory 106. In one embodiment herein, the memory 106 incorporates at least one storage medium selected from a flash memory, a hard disk storage medium, a multimedia card storage medium, a card-type memory, random-access memory (RAM), static random-access memory (SRAM), read-only memory (ROM), electrically erasable programmable read-only memory (EEPROM), programmable read-only memory (PROM), magnetic memory, magnetic disk storage, and optical disk storage. The memory 106 stores the one or more instructions executable by the at least one processor 104 for operation of the plurality of modules 116.

[0055] In one embodiment herein, the memory 106 stores executable instructions associated with plurality of modules 116. The at least one processor 104 retrieves the executable instructions from the memory 106 and executes the plurality of modules 116 for identity generation, localization, verification, and recovery of the laundry bag based on visual and physical characteristics associated with the laundry bag. The plurality of modules 116 incorporates a registration module 118, a feature acquisition module 120, an artificial intelligence (AI) identity generation module 122, a centralized database module 124, an AI-based rack localization module 126, a user interaction module 128, a collection verification module 130, and a recovery module 132.

[0056] In one embodiment herein, the registration module 118 associates the laundry bag with a corresponding user identity based on at least one of a registration number, an identity card, and authentication through a user device. Association between the laundry bag and the corresponding user identity establishes a user-linked record for subsequent identity generation, localization, verification, and recovery operations performed by the system 100.

[0057] In one embodiment herein, the feature acquisition module 120 receives visual and physical characteristics of the laundry bag acquired through one or more cameras and at least one weight sensor communicatively coupled to the computing device 102. The visual and physical characteristics comprise image appearance, colour distribution, shape, dimensions, texture characteristics, and weight. The feature acquisition module 120 provides the acquired

visual and physical characteristics to the artificial intelligence (AI) identity generation module 122 for generation of a Laundry Identity Profile associated with the laundry bag.

[0058] In one embodiment herein, the feature acquisition module 120 preprocesses the visual and physical characteristics acquired through the one or more cameras and the at least one weight sensor and converts the visual and physical characteristics into machine-readable feature representations corresponding to image appearance, colour distribution, shape, dimensions, texture characteristics, and weight of the laundry bag. The machine-readable feature representations are provided to the artificial intelligence (AI) identity generation module 122 for feature extraction, feature embedding generation, similarity encoding, and generation of the Laundry Identity Profile associated with the laundry bag.

[0059] In one embodiment herein, the artificial intelligence (AI) identity generation module 122 extracts features from the machine-readable feature representations received from the feature acquisition module 120 and converts the extracted features into numerical feature vectors. The artificial intelligence (AI) identity generation module 122 generates the Laundry Identity Profile from the numerical feature vectors and associates the Laundry Identity Profile with the corresponding user identity. The Laundry Identity Profile thereby provides a feature-based digital representation of the physical laundry bag for subsequent localization, collection verification, re-identification, and recovery operations performed by the system 100.

[0060] In one embodiment herein, the artificial intelligence (AI) identity generation module 122 performs feature extraction, feature embedding generation, similarity encoding, and identity profile generation using the visual and physical characteristics comprising image appearance, colour distribution, shape, dimensions, texture characteristics, and weight. The feature embedding generation produces numerical feature vectors representing the extracted characteristics, and the similarity encoding enables comparison of the numerical feature representations with stored Laundry Identity Profiles during rack localization, collection verification, and recovery.

[0061] In one embodiment herein, the AI-based rack localization module 126 processes the rack image data through a multi-object detection model configured to detect a plurality of individual laundry bags present within a common field of view of the one or more overhead cameras. The multi-object detection model may comprise a YOLO-based object-detection

model or another suitable computer-vision-based object-detection model. Each detected laundry bag is spatially isolated from the rack image data based on a corresponding detected image region and is individually processed for extraction of visible characteristics and subsequent similarity matching with the stored Laundry Identity Profiles.

[0062] In one embodiment herein, the artificial intelligence (AI) identity generation module 122 utilizes a deep-learning-based visual feature extractor to generate a numerical feature embedding representing an overall image appearance of the laundry bag. During rack localization, collection verification, and recovery, a corresponding numerical feature embedding generated from a detected laundry bag is compared with a numerical feature embedding represented in the Laundry Identity Profile. The numerical feature embeddings thereby provide a machine-processable representation of the overall appearance of the laundry bag for determining correspondence between a detected physical laundry bag and the Laundry Identity Profile associated with the corresponding user identity.

[0063] In one embodiment herein, each characteristic extracted from a detected laundry bag is converted into a corresponding normalized similarity value ranging from 0 to 1, wherein a similarity value approaching 1 represents greater correspondence with a corresponding characteristic represented in the Laundry Identity Profile and a similarity value approaching 0 represents lower correspondence. The normalized similarity values comprise an image-appearance similarity, a colour similarity, a shape similarity, a dimensional similarity, a texture similarity, and, where a weight measurement is available, a weight similarity.

[0064] In one embodiment herein, the colour similarity is determined by comparing a colour distribution extracted from the detected laundry bag with a colour distribution represented in the Laundry Identity Profile. A colour histogram or an equivalent colour representation may be generated for the detected laundry bag and compared with the corresponding stored colour representation to determine the colour similarity. The colour similarity thereby represents correspondence between dominant and distributed colour characteristics of the detected laundry bag and the previously acquired laundry bag characteristics.

[0065] In one embodiment herein, the shape similarity is determined based on comparison of at least one of an outline, contour, aspect ratio, and general geometric characteristic extracted from the detected laundry bag with a corresponding shape representation stored

in the Laundry Identity Profile. The dimensional similarity is determined by estimating at least one of a height, width, and approximate size of the detected laundry bag and comparing the estimated dimensions with corresponding dimensions represented in the Laundry Identity Profile. Differences between the estimated dimensions and the stored dimensions are normalized so that dimensional variations caused by folding, compression, deformation, or changes in contents of the laundry bag do not independently result in rejection of the detected laundry bag.

[0066] In one exemplary implementation herein, dimensions represented in the Laundry Identity Profile may correspond to approximately 60 cm by 35 cm, while dimensions estimated from a subsequently detected laundry bag may correspond to approximately 58 cm by 34 cm. The artificial intelligence (AI) identity generation module 122 may determine a dimensional similarity of approximately 0.90 based on the comparison. The dimensional similarity is determined so as to accommodate limited dimensional variations resulting from folding, compression, deformation, variation in orientation, or variation in contents of the laundry bag without independently rejecting the detected laundry bag.

[0067] In one embodiment herein, the texture similarity is determined from at least one of a fabric pattern, texture distribution, surface roughness representation, and another distinguishable surface characteristic extracted from the detected laundry bag and compared with a corresponding texture representation stored in the Laundry Identity Profile. The image-appearance similarity is determined by comparing the numerical feature embedding generated from the detected laundry bag with the corresponding numerical feature embedding stored in the Laundry Identity Profile. In one embodiment herein, cosine similarity is employed for comparing the numerical feature embeddings, although another suitable vector-similarity measure may be employed.

[0068] In one exemplary implementation herein, where a texture representation stored in the Laundry Identity Profile represents a coarse woven surface characteristic and a subsequently detected laundry bag exhibits a corresponding coarse woven texture distribution, the artificial intelligence (AI) identity generation module 122 may determine a texture similarity of approximately 0.84. The texture similarity is evaluated together with

image-appearance similarity, colour similarity, shape similarity, and dimensional similarity and is not independently determinative of identification of the laundry bag.

[0069] In one embodiment herein, the artificial intelligence (AI) identity generation module 122 determines the feature similarity score based on respective normalized similarity values associated with the extracted characteristics and respective weighting factors assigned thereto. The weighting factors establish a relative contribution of image appearance, colour distribution, shape, dimensions, and texture characteristics during comparison of a detected laundry bag with a corresponding Laundry Identity Profile. In a preferred initial configuration, image-appearance similarity is assigned a greater contribution than the individual colour, shape, texture, and dimensional similarities, thereby allowing the overall visual appearance of the laundry bag to provide a primary identification characteristic while the remaining characteristics provide complementary identification information.

[0070] In a preferred initial configuration, respective weighting factors comprise 0.40 for image-appearance similarity, 0.20 for colour similarity, 0.15 for shape similarity, 0.10 for texture similarity, 0.05 for dimensional similarity, and 0.10 for weight similarity. The weighting factors collectively sum to 1.00. Accordingly, image appearance provides a primary contribution to identification, while colour, shape, texture, dimensions, and weight provide supporting identification information. In the preferred initial configuration, an overall confidence score C is represented according to:

$$C = 0.40S_a + 0.20S_c + 0.15S_s + 0.10S_t + 0.05S_d + 0.10S_w$$

wherein S_a represents the image-appearance similarity, S_c represents the colour similarity, S_s represents the shape similarity, S_t represents the texture similarity, S_d represents the dimensional similarity, and S_w represents the weight similarity. Each similarity value is normalized between 0 and 1. The weighting configuration represents a preferred initial implementation and may be calibrated according to experimentally collected laundry-bag data.

[0071] In one embodiment herein, the respective similarity values are evaluated collectively such that failure of an individual characteristic to provide a perfect correspondence does not independently result in rejection of a candidate laundry bag. The multimodal similarity comparison accommodates variations in appearance resulting from at least one of rotation,

folding, compression, deformation, partial occlusion, variation in illumination, and variation in quantities of clothing contained within the laundry bag. The combined comparison thereby provides improved discrimination between candidate laundry bags relative to identification based on a single visible or physical characteristic.

[0072] The weighting factors and decision thresholds may be calibrated or optimized using laundry-bag image and sensor data acquired under different orientations, illumination conditions, folding conditions, occlusion levels, deformation conditions, and filling conditions. Weight is treated as a supporting verification characteristic because a weight of the laundry bag may vary between submission and collection as a result of laundry processing or changes in contents, while visual appearance characteristics provide the primary information for rack localization and rack-wide recovery.

[0073] In one embodiment herein, the feature similarity score determined by the artificial intelligence (AI) identity generation module 122 is utilized by the AI-based rack localization module 126 for similarity matching between detected laundry bags and stored Laundry Identity Profiles. The collection verification module 130 further utilizes the feature similarity score together with a weight similarity score for generating a confidence score, and the recovery module 132 utilizes similarity and confidence information for re-identification of a misplaced laundry bag from candidate laundry bags detected across the storage racks.

[0074] In one embodiment herein, the centralized database module 124 stores the Laundry Identity Profile in association with the corresponding user identity, laundry-processing status, rack-location information, collection history, and verification and recovery records. The centralized database module 124 synchronizes the Laundry Identity Profile, the laundry-processing status, the rack-location information, the collection history, and the verification and recovery records with the AI-based rack localization module 126 and the user interaction module 128.

[0075] In one embodiment herein, the AI-based rack localization module 126 is operatively connected to the centralized database module 124 and receives rack image data acquired by one or more overhead cameras arranged to continuously monitor respective sections of storage racks containing a plurality of laundry bags positioned without a predefined sorting sequence. The AI-based rack localization module 126 performs multi-object laundry bag

detection on the rack image data, extracts visible characteristics comprising colour distribution, shape, dimensions, texture characteristics, and image appearance from detected laundry bags, and compares the visible characteristics with stored Laundry Identity Profiles using similarity matching.

[0076] In a preferred embodiment herein, the one or more overhead cameras comprise fixed RGB cameras mounted above respective sections of the storage racks. Each overhead camera preferably provides a resolution of 1080p or greater and is positioned to cover a defined rack section containing a plurality of laundry bags. The one or more overhead cameras may continuously or periodically acquire the rack image data according to an operating configuration of the system 100.

[0077] In one embodiment herein, adjacent overhead cameras are positioned to provide overlapping fields of view. In a preferred configuration, the overlapping fields of view are approximately 10% to 20% between adjacent camera coverage areas. The overlapping fields of view reduce loss of detectable laundry-bag characteristics resulting from partial occlusion, overlapping laundry bags, or positioning of a laundry bag near a boundary between adjacent camera coverage areas.

[0078] In one embodiment herein, the storage racks and the corresponding fields of view of the one or more overhead cameras are spatially calibrated such that image coordinates associated with a detected laundry bag are mapped to a corresponding rack number, shelf number, and slot position. Based on the spatial calibration, the AI-based rack localization module 126 converts a detected image position of the identified laundry bag into the rack-location information comprising the rack number, the shelf number, and the slot position.

[0079] In one embodiment herein, the AI-based rack localization module 126 identifies a detected laundry bag based on the highest similarity match between the visible characteristics of the detected laundry bag and the stored Laundry Identity Profiles. Following identification, the AI-based rack localization module 126 determines a rack number, a shelf number, and a slot position corresponding to the identified laundry bag. The rack number, shelf number, and slot position collectively establish rack-location information associated with the identified laundry bag, and the centralized database module 124 stores the

determined rack-location information for subsequent collection verification and recovery operations.

[0080] In one embodiment herein, the user interaction module 128 receives the rack number, the shelf number, and the slot position determined by the AI-based rack localization module 126. The user interaction module 128 provides laundry-processing status, collection availability status, and location information associated with the laundry bag to the corresponding user through the user device. The user interaction module 128 further provides laundry submission confirmation, laundry-completion notification, the rack number, the shelf number, the slot position, and collection availability status associated with the laundry bag.

[0081] In one embodiment herein, the collection verification module 130 compares current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity. The collection verification module 130 determines a feature similarity score and a weight similarity score and generates a confidence score based on the feature similarity score and the weight similarity score. The collection verification module 130 authorizes collection of the retrieved laundry bag when the confidence score satisfies a predefined threshold.

[0082] In one embodiment herein, the collection verification module 130 determines the weight similarity score based on a difference between an originally recorded weight of the laundry bag and a current measured weight of the retrieved laundry bag relative to the originally recorded weight. The collection verification module 130 generates the confidence score by combining the feature similarity score and the weight similarity score according to respective weighting constants. Upon authorization of collection based on satisfaction of the predefined threshold, the collection verification module 130 updates a collection status associated with the laundry bag through the centralized database module 124.

[0083] In a preferred initial configuration, the predefined threshold applied by the collection verification module 130 for authorizing collection is 0.85. The collection verification module 130 further compares a highest confidence score associated with a candidate laundry bag with a second-highest confidence score associated with another candidate laundry bag to determine whether sufficient discrimination exists between the candidate laundry bags.

[0084] In the preferred initial configuration, collection is authorized when the highest confidence score is at least 0.85 and a difference between the highest confidence score and the second-highest confidence score is at least 0.08. When the highest confidence score fails to satisfy the predefined threshold or the difference between the highest confidence score and the second-highest confidence score is less than the required candidate-separation threshold, the collection verification module 130 identifies the verification result as ambiguous or potentially incorrect and initiates additional verification or provides the mismatch condition to the recovery module 132.

[0085] The confidence threshold and the candidate-separation threshold represent preferred initial configuration values and are not limited to fixed values. The confidence threshold and the candidate-separation threshold may be calibrated based on experimentally acquired laundry-bag data so as to reduce incorrect authorization where two or more candidate laundry bags exhibit visually similar characteristics.

[0086] In one embodiment herein, failure of the confidence score associated with the retrieved laundry bag to satisfy the predefined threshold establishes a mismatch condition for initiating a recovery operation. The recovery module 132 further initiates the recovery operation upon detection of absence of the laundry bag from the determined rack location. In response to the mismatch condition or absence condition, the recovery module 132 initiates rack-wide scanning, and the one or more overhead cameras acquire rack image data corresponding to candidate laundry bags positioned across the storage racks.

[0087] In one embodiment herein, the recovery module 132 compares the candidate laundry bags detected from the rack image data with the stored Laundry Identity Profile using similarity matching and determines respective confidence scores associated with the candidate laundry bags. The recovery module 132 selects the candidate laundry bag having the highest confidence score as the most probable matching laundry bag, determines a corrected rack number, shelf number, and slot position corresponding to the selected candidate laundry bag, and updates the corrected rack-location information through the centralized database module 124. The corrected rack-location information is provided to the corresponding user through the user interaction module 128 for subsequent retrieval and

verification of the laundry bag. The recovery module 132 further initiates recovery in response to a missing-bag report received through the user device.

[0088] In one embodiment herein, during rack-wide recovery, the recovery module 132 evaluates candidate laundry bags according to their respective confidence scores and retains candidate laundry bags satisfying a recovery confidence threshold for further analysis. In a preferred initial recovery configuration, candidate laundry bags having confidence scores of approximately 0.70 or greater are retained and ranked according to their respective confidence scores.

[0089] The recovery module 132 identifies the retained candidate laundry bag having the highest valid confidence score as the most probable matching laundry bag and determines corrected rack-location information corresponding thereto. When no candidate laundry bag satisfies the recovery confidence threshold, the recovery module 132 records the laundry bag as unresolved and initiates at least one of additional rack-wide scanning and operator intervention.

[0090] The recovery confidence threshold represents a preferred initial configuration and may be calibrated based on image data acquired under different storage-rack conditions. Selection of a recovery confidence threshold lower than the collection authorization threshold permits potentially matching candidate laundry bags to be retained for further analysis without independently authorizing final collection thereof.

[0091] As used herein, autonomous recovery refers to automatic detection, rack-wide searching, re-identification, and localization of a misplaced laundry bag by the system 100. Autonomous recovery does not require or denote physical movement of the misplaced laundry bag by the system 100. Upon re-identification of the misplaced laundry bag, the recovery module 132 automatically determines corrected rack-location information, the centralized database module 124 updates the corrected rack-location information, and the user interaction module 128 communicates the corrected rack-location information to the corresponding user through the user device for subsequent physical retrieval of the laundry bag.

[0092] In one embodiment herein, the artificial intelligence (AI) identity generation module (122), the centralized database module 124, the AI-based rack localization module 126, the collection verification module 130, and the recovery module 132 operate cooperatively with the one or more cameras and the at least one weight sensor to maintain correspondence between the physical laundry bag, the Laundry Identity Profile associated with the laundry bag, and the determined rack-location information. The system 100 thereby performs identity generation from acquired visual and physical characteristics, localization of laundry bags positioned without the predefined sorting sequence, verification of a retrieved physical laundry bag, and re-identification and corrected localization following detection of a mismatch or absence condition, independently of an RFID tag, a barcode label, a QR code, and other physical identifiers associated with the laundry bag.

[0093] In a preferred sensing configuration, the at least one weight sensor comprises a digital weighing platform incorporating one or more load-cell sensors positioned at a laundry submission station and a laundry collection station. The weighing platform is dimensioned to receive the laundry bag in a closed condition and may have a platform dimension of approximately 40 cm to 60 cm by approximately 40 cm to 60 cm. The one or more load-cell sensors generate a weight measurement and communicate the weight measurement to the computing device 102 through a corresponding sensor interface.

[0094] During submission of the laundry bag, the laundry bag is positioned substantially completely on the weighing platform while the one or more cameras acquire the visual characteristics of the laundry bag. The measured initial weight and the acquired visual characteristics are associated with the corresponding user identity and processed for generation of the Laundry Identity Profile. Accordingly, submission of the laundry bag preferably utilizes both camera-derived characteristics and a weight measurement.

[0095] In one embodiment herein, during submission of the laundry bag, the one or more cameras acquire one or more images of the laundry bag from a controlled image-acquisition position while the laundry bag is positioned substantially completely on the weighing platform. The controlled image-acquisition position provides a repeatable imaging relationship between the laundry bag and the one or more cameras for acquiring image appearance, colour distribution, shape, dimensions, and texture characteristics. The feature

acquisition module 120 receives the acquired visual characteristics together with the initial weight measurement for generation of the Laundry Identity Profile associated with the corresponding user identity.

[0096] Following completion of laundry processing, the laundry bag may be positioned at any available position on the storage racks without a predefined sorting sequence. During post-laundry rack localization, the one or more overhead cameras provide a primary sensing arrangement for detecting the laundry bags, extracting the visible characteristics, performing similarity matching, and determining the rack-location information. A weight sensor is not required at each rack position, thereby avoiding installation of respective weighing sensors throughout the storage racks.

[0097] In one embodiment herein, the overhead-camera arrangement enables the AI-based rack localization module 126 to identify and localize a plurality of laundry bags positioned within respective rack sections without requiring each laundry bag to be individually presented in front of a camera or manually arranged according to the corresponding user identity. The AI-based rack localization module 126 performs localization after placement of the laundry bags by processing the rack image data, detecting the plurality of laundry bags within respective camera fields of view, extracting visible characteristics of the detected laundry bags, and comparing the visible characteristics with the stored Laundry Identity Profiles.

[0098] During collection verification, the retrieved laundry bag is presented at the collection station and positioned on a calibrated weighing platform, wherein the one or more cameras acquire current visual characteristics and the at least one weight sensor measures a current weight of the retrieved laundry bag. The collection verification module 130 compares the current visual characteristics and the current weight with corresponding characteristics represented in the Laundry Identity Profile and determines respective similarity values therefrom. The collection verification module 130 utilizes the respective similarity values for determining the confidence score and verifying whether the retrieved laundry bag corresponds to the Laundry Identity Profile associated with the corresponding user identity.

[0099] During misplaced-bag recovery, rack-wide searching and re-identification are performed primarily using the rack image data acquired by the one or more overhead

cameras. Weight measurement is not required during image-based searching across the storage racks. After a probable matching laundry bag is physically retrieved from a corrected rack location, the current weight of the probable matching laundry bag may be measured at the collection station and utilized as an additional verification signal before final collection authorization.

[00100] In one embodiment herein, the sensing configuration of the system 100 comprises a submission configuration, a post-laundry localization configuration, a collection configuration, and a misplaced-bag recovery configuration. In the submission configuration, the one or more cameras and the at least one weight sensor acquire visual and physical characteristics for generation of the Laundry Identity Profile. In the post-laundry localization configuration, the one or more overhead cameras acquire rack image data for identification of the laundry bag and determination of the rack number, shelf number, and slot position. In the collection configuration, the one or more cameras and the at least one weight sensor acquire current visual and physical characteristics for verification of the retrieved laundry bag. In the misplaced-bag recovery configuration, the one or more overhead cameras acquire rack-wide image data for re-identification of the laundry bag and determination of corrected rack-location information.

[00101] In one embodiment herein, the sensing and processing configuration of the system 100 is scalable across a plurality of storage-rack sections containing a plurality of randomly positioned laundry bags. The one or more overhead cameras provide respective camera coverage areas for detecting and identifying laundry bags without requiring a respective physical identifier or weight sensor to be installed for each laundry bag or rack position. Accordingly, AI-based visual identification and localization provide the primary mechanism for post-laundry localization and misplaced-bag recovery, while weight measurement provides a supporting physical verification characteristic during submission and collection.

[00102] In one exemplary implementation herein, the system 100 is deployable in a large-scale laundry environment containing hundreds or thousands of laundry bags distributed across a plurality of storage racks. The respective storage-rack sections are monitored through the one or more overhead cameras, and the AI-based rack localization module 126 processes the corresponding rack image data for multi-object laundry bag detection,

similarity matching, and rack-location determination without requiring predetermined user-identity-based sorting of the laundry bags.

[00103] In one exemplary embodiment herein, the laundry bag associated with registration number 23MIS0050 is submitted for laundry processing. The registration module 118 associates the laundry bag with the corresponding user identity. The feature acquisition module 120 acquires visual and physical characteristics of the laundry bag through the one or more cameras and the at least one weight sensor. The acquired visual and physical characteristics correspond to a dark blue colour distribution, cylindrical shape, textured fabric appearance, and a recorded weight of 4.8 kg. The artificial intelligence (AI) identity generation module 122 processes the acquired visual and physical characteristics, generates numerical feature vectors, and generates the Laundry Identity Profile associated with the corresponding user identity. The centralized database module 124 stores the Laundry Identity Profile together with the corresponding user identity and laundry-processing status.

[00104] Following laundry processing, the processed laundry bag is positioned on the storage racks without the predefined sorting sequence. In one embodiment herein, the laundry bag is positioned at Rack B, Shelf 3, Slot 12. The one or more overhead cameras acquire rack image data from respective sections of the storage racks. The AI-based rack localization module 126 performs multi-object laundry bag detection, extracts visible characteristics of detected laundry bags, compares the visible characteristics with stored Laundry Identity Profiles using similarity matching, and identifies the laundry bag based on the highest similarity match.

[00105] The AI-based rack localization module 126 determines Rack B, Shelf 3, Slot 12 as the rack-location information associated with the identified laundry bag. The centralized database module 124 stores the rack-location information, and the user interaction module 128 provides the rack-location information and collection availability status to the corresponding user through the user device. During collection, the collection verification module 130 compares current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity. In one embodiment herein, the retrieved laundry bag presents a dark blue appearance and a current measured weight of 4.7 kg. The collection verification module 130 determines the feature

similarity score and the weight similarity score, generates the confidence score, and authorizes collection when the confidence score satisfies the predefined threshold.

[00106] In one exemplary implementation, comparison of the retrieved laundry bag with the corresponding Laundry Identity Profile provides an image-appearance similarity of 0.94, a colour similarity of 0.92, a shape similarity of 0.88, a texture similarity of 0.84, a dimensional similarity of 0.90, and a weight similarity of 0.95. Applying the preferred respective weighting factors produces an overall confidence score of approximately 0.91. The resulting confidence score represents a strong candidate correspondence and may satisfy the preferred collection authorization threshold when the corresponding candidate-separation requirement is also satisfied.

[00107] Upon failure of the retrieved laundry bag to satisfy the predefined threshold or absence of the laundry bag from the determined rack location, the recovery module 132 initiates rack-wide scanning of the storage racks. In one embodiment herein, similarity matching identifies the candidate laundry bag having matching dark blue colour, texture characteristics, and dimensions at Rack D, Shelf 2, Slot 7. The recovery module 132 selects the candidate laundry bag having the highest confidence score and determines Rack D, Shelf 2, Slot 7 as corrected rack-location information.

[00108] The recovery module 132 updates the corrected rack-location information through the centralized database module 124, and the user interaction module 128 provides the corrected rack-location information to the corresponding user through the user device for subsequent collection of the laundry bag. The exemplary working operation demonstrates tokenless identity generation, localization, collection verification, re-identification, and recovery based on visual and physical characteristics associated with the laundry bag.

[00109] According to an exemplary embodiment of the invention, FIG. 2 refers to a functional block diagram of a system 100 for tokenless laundry bag identity generation, localization, verification, and recovery. In one embodiment herein, the functional block diagram 200 depicts a sequential operational flow from association of the laundry bag with the corresponding user identity through acquisition of visual and physical characteristics, generation of the Laundry Identity Profile, laundry processing, rack placement, rack localization, collection verification, recovery, and corrected rack-location updating.

[00110] At step 202, the registration module 118 associates the laundry bag with the corresponding user identity based on at least one of the registration number, the identity card, and authentication through the user device. Association of the laundry bag with the corresponding user identity establishes the user-linked record for subsequent identity generation, localization, verification, and recovery operations. At step 204, the feature acquisition module 120 acquires visual and physical characteristics of the laundry bag through the one or more cameras and the at least one weight sensor. The visual and physical characteristics are image appearance, colour distribution, shape, dimensions, texture characteristics, and weight. The feature acquisition module 120 transfers the acquired visual and physical characteristics for generation of the Laundry Identity Profile associated with the laundry bag.

[00111] At step 206, the artificial intelligence (AI) identity generation module 122 extract features from the visual and physical characteristics and converts the extracted features into numerical feature vectors. The artificial intelligence (AI) identity generation module 122 generates the Laundry Identity Profile from the numerical feature vectors and associates the Laundry Identity Profile with the corresponding user identity. The Laundry Identity Profile provides a feature-based representation of the laundry bag for identity generation, localization, verification, and recovery operations.

[00112] In one embodiment herein, the artificial intelligence (AI) identity generation module 122 determines the feature similarity score from respective normalized similarity values associated with the extracted visual characteristics and respective weighting factors assigned thereto. The feature similarity score is represented according to:

$$S_f = \sum_{i=1}^{n} w_i \times f_i$$

wherein S_f represents the feature similarity score, w_i represents a weighting factor assigned to a corresponding extracted visual characteristic, and f_i represents a normalized similarity value associated with the corresponding extracted visual characteristic. The normalized similarity values comprise image-appearance similarity, colour similarity, shape similarity, dimensional similarity, and texture similarity. The respective weighting factors establish the relative contribution of the extracted visual characteristics during comparison of a detected

laundry bag with a stored Laundry Identity Profile. In a preferred configuration, the image-appearance similarity is assigned a greater contribution than the individual colour, shape, texture, and dimensional similarities.

[00113] In a preferred initial configuration, the collection verification module 130 determines an overall confidence score based on the image-appearance similarity, colour similarity, shape similarity, texture similarity, dimensional similarity, and weight similarity. The overall confidence score C is represented according to:

$$C = 0.40S_a + 0.20S_c + 0.15S_s + 0.10S_t + 0.05S_d + 0.10S_w$$

wherein S_a represents the image-appearance similarity, S_c represents the colour similarity, S_s represents the shape similarity, S_t represents the texture similarity, S_d represents the dimensional similarity, and S_w represents the weight similarity, each normalized between 0 and 1. In the preferred initial configuration, the image-appearance similarity is assigned a weighting factor of 0.40, the colour similarity is assigned a weighting factor of 0.20, the shape similarity is assigned a weighting factor of 0.15, the texture similarity is assigned a weighting factor of 0.10, the dimensional similarity is assigned a weighting factor of 0.05, and the weight similarity is assigned a weighting factor of 0.10. The respective weighting factors may be calibrated or optimized using experimentally collected laundry-bag data under different orientations, lighting conditions, folding conditions, occlusion levels, and filling conditions.

[00114] At step 208, the centralized database module 124 stores the Laundry Identity Profile in association with the corresponding user identity, laundry-processing status, rack-location information, collection history, and verification and recovery records. The centralized database module 124 provides the stored Laundry Identity Profile for subsequent comparison and localization operations performed by the AI-based rack localization module 126 and for verification operations performed by the collection verification module 130. At step 210, the laundry bag undergoes laundry processing through washing, drying, and quality checking. Completion of the laundry processing establishes a processed laundry bag for subsequent placement on the storage racks.

[00115] At step 212, the processed laundry bag is positioned on the storage racks without the predefined sorting sequence. Placement of processed laundry bags without the predefined sorting sequence permits storage of plurality of laundry bags without user-identity-based rack

organization. The rack-location information is determined after placement through image-based localization rather than through a predetermined rack assignment.

[00116] At step 214, the one or more overhead cameras continuously acquire rack image data from respective sections of the storage racks. The rack image data represents plurality of laundry bags positioned on the storage racks without the predefined sorting sequence. The rack image data is transferred to the AI-based rack localization module 126 for multi-object laundry bag detection, feature extraction, similarity matching, and location determination. At step 216, the AI-based rack localization module 126 performs multi-object laundry bag detection on the rack image data and extracts visible characteristics of detected laundry bags. The visible characteristics are colour distribution, shape, dimensions, texture characteristics, and image appearance. The AI-based rack localization module 126 compares the visible characteristics with stored Laundry Identity Profiles using similarity matching and identifies the laundry bag based on the highest similarity match.

[00117] At step 216, the AI-based rack localization module 126 further determines the rack number, the shelf number, and the slot position corresponding to the identified laundry bag. A localized position of the laundry bag is represented according to:

$$L_b = (R_i, S_j, P_k)$$

L_b represents the localized position of the laundry bag, R_i represents the rack number, S_j represents the shelf number, and P_k represents the slot position. The determined rack number, the shelf number, and the slot position establish the rack-location information associated with the identified laundry bag.

[00118] At step 218, the user interaction module 128 receives the rack number, the shelf number, and the slot position determined by the AI-based rack localization module 126. The user interaction module 128 provides laundry-processing status, collection availability status, and location information associated with the laundry bag to the corresponding user through the user device. The location information directs the corresponding user to the determined rack number, the shelf number, and the slot position for collection of the laundry bag.

[00119] At step 220, the collection verification module 130 compares current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile

associated with the corresponding user identity. The collection verification module 130 determines the feature similarity score and the weight similarity score and generates the confidence score based on the feature similarity score and the weight similarity score. In one embodiment herein, the collection verification module 130 determines the weight similarity score according to:

$$W_s = 1 - |W_o - W_c| / W_o$$

W_s represents the weight similarity score, W_o represents an originally recorded weight of the laundry bag, and W_c represents a current measured weight of the retrieved laundry bag.

[00120] The collection verification module 130 generates the confidence score according to:

$$C = \alpha S_f + \beta W_s$$

C represents the confidence score, S_f represents the feature similarity score, W_s represents the weight similarity score, and α and β represent respective weighting constants. The collection verification module 130 authorizes collection of the retrieved laundry bag when the confidence score satisfies the predefined threshold. Upon authorization of collection, the collection verification module 130 updates the collection status associated with the laundry bag through the centralized database module 124.

[00121] In a preferred initial configuration, the collection verification module 130 determines the confidence score based on a weighted combination of the image-appearance similarity, colour similarity, shape similarity, texture similarity, dimensional similarity, and weight similarity. The preferred confidence score is represented according to

$$C = 0.40S_a + 0.20S_c + 0.15S_s + 0.10S_t + 0.05S_d + 0.10S_w$$

wherein S_a represents the image-appearance similarity, S_c represents the colour similarity, S_s represents the shape similarity, S_t represents the texture similarity, S_d represents the dimensional similarity, and S_w represents the weight similarity.

[00122] In one embodiment herein, the collection verification module 130 ranks candidate laundry bags according to their respective confidence scores. In the preferred initial configuration, the collection verification module 130 authorizes collection when a highest

confidence score is at least 0.85 and a difference between the highest confidence score and a second-highest confidence score is at least 0.08. Failure to satisfy either condition establishes an ambiguous or mismatch condition for additional verification or recovery.

[00123] At step 222, the recovery module 132 initiates rack-wide scanning of the storage racks upon detection of the at least one condition selected from absence of the laundry bag from the determined rack location and failure of the confidence score associated with the retrieved laundry bag to satisfy the predefined threshold. The one or more overhead cameras continuously scan the storage racks during the rack-wide scanning and acquire rack image data associated with candidate laundry bags. At step 222, the recovery module 132 compares the candidate laundry bags detected during the rack-wide scanning with the stored Laundry Identity Profile using similarity matching. The recovery module 132 determines respective confidence scores associated with the candidate laundry bags and selects the candidate laundry bag having the highest confidence score. Selection of the candidate laundry bag is represented according to:

$$M_b = \max(C_i)$$

M_b represents the selected candidate laundry bag and C_i represents the confidence score associated with the candidate laundry bag. The recovery module 132 determines a corrected rack number, the shelf number, and the slot position of the selected candidate laundry bag.

[00124] During recovery, the recovery module 132 may retain candidate laundry bags having a confidence score of approximately 0.70 or greater and rank the retained candidate laundry bags according to their respective confidence scores. The retained candidate laundry bag having the highest valid confidence score is selected as the most probable matching laundry bag. When no candidate laundry bag satisfies the recovery criterion, the recovery module 132 records an unresolved condition and initiates additional rack scanning or operator intervention.

[00125] At step 224, the corrected rack-location information is updated through the centralized database module 124. The user interaction module 128 provides the corrected rack-location information to the corresponding user through the user device for subsequent

collection of the laundry bag. The updated information identifies the corrected rack number, the shelf number, and the slot position associated with the selected candidate laundry bag.

[00126] According to an exemplary embodiment of the invention, FIG. 3 refers to a process flow diagram of the system 100 for tokenless laundry bag identity generation, localization, verification, and recovery. In one embodiment herein, the process flow diagram 300 illustrates sequential processing from association of the laundry bag with the corresponding user identity through feature acquisition, Laundry Identity Profile generation, laundry processing, rack localization, collection verification, recovery, and corrected rack-location updating.

[00127] At step 302, the registration module 118 associates the laundry bag with the corresponding user identity based on at least one of the registration number, the identity card, and authentication through the user device. The association establishes an association between the laundry bag and the corresponding user identity for subsequent processing of the laundry bag through the system 100. At step 304, the feature acquisition module 120 acquires visual and physical characteristics of the laundry bag through the one or more cameras and the at least one weight sensor. The visual and physical characteristics are image appearance, colour distribution, shape, dimensions, texture characteristics, and weight. The acquired visual and physical characteristics provide input information for generation of the Laundry Identity Profile associated with the laundry bag.

[00128] At step 306, the artificial intelligence (AI) identity generation module 122 extract features from the visual and physical characteristics and converts the extracted features into numerical feature vectors. The artificial intelligence (AI) identity generation module 122 generates the Laundry Identity Profile from the numerical feature vectors and associates the Laundry Identity Profile with the corresponding user identity. The Laundry Identity Profile provides the feature-based representation of the laundry bag for identification, localization, verification, and recovery operations. At step 308, the centralized database module 124 stores the Laundry Identity Profile in association with the corresponding user identity, laundry-processing status, rack-location information, collection history, and verification and recovery records. The stored Laundry Identity Profile provides reference information for similarity matching performed during rack localization, collection verification, and recovery.

[00129] At step 310, the laundry bag undergoes laundry processing through washing, drying, and quality checking. Completion of the laundry processing establishes the laundry bag for placement on storage racks and subsequent localization. At step 312, the processed laundry bag is positioned on the storage racks without the predefined sorting sequence. The storage racks contain the plurality of laundry bags positioned independently of user-identity-based rack organization. The physical location of the laundry bag is subsequently determined through rack image data and similarity matching.

[00130] At step 314, the one or more overhead cameras continuously acquire rack image data from respective sections of the storage racks. The rack image data represents the laundry bags positioned across the storage racks and provides image information for subsequent multi-object laundry bag detection and localization. At step 316, the AI-based rack localization module 126 receives the rack image data and performs multi-object laundry bag detection. The AI-based rack localization module 126 extracts visible characteristics of detected laundry bags and compares the visible characteristics with stored Laundry Identity Profiles using similarity matching. Based on the highest similarity match, the AI-based rack localization module 126 identifies the laundry bag and determines the rack number, the shelf number, and the slot position corresponding to the identified laundry bag.

[00131] At step 318, the user interaction module 128 receives the rack number, the shelf number, and the slot position determined by the AI-based rack localization module 126. The user interaction module 128 provides laundry-processing status, collection availability status, and location information associated with the identified laundry bag to the corresponding user through the user device. The location information directs the corresponding user to the rack number, the shelf number, and the slot position associated with the identified laundry bag.

[00132] At step 320, the collection verification module 130 verifies the retrieved laundry bag by comparing current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity. The collection verification module 130 determines the feature similarity score and the weight similarity score and generates the confidence score based on the feature similarity score and the weight similarity score. At step 322, the collection verification module 130 evaluates the confidence score relative to the predefined threshold. Satisfaction of the predefined threshold

establishes the match between the retrieved laundry bag and the Laundry Identity Profile associated with the corresponding user identity. Failure of the confidence score to satisfy the predefined threshold establishes a mismatch condition for initiation of recovery.

[00133] At step 324, upon satisfaction of the predefined threshold, the collection verification module 130 authorizes collection of the retrieved laundry bag. The centralized database module 124 updates the collection status associated with the laundry bag to record completion of collection. At step 326, upon failure of the confidence score to satisfy the predefined threshold, the recovery module 132 initiates rack-wide scanning of the storage racks. The recovery module 132 also initiates rack-wide scanning upon absence of the laundry bag from the determined rack location. The one or more overhead cameras acquire rack image data from the storage racks, and the recovery module 132 compares candidate laundry bags detected during the rack-wide scanning with the stored Laundry Identity Profile using similarity matching.

[00134] At step 326, the recovery module 132 determines respective confidence scores associated with the candidate laundry bags and selects the candidate laundry bag having the highest confidence score. The selected candidate laundry bag represents the matching laundry bag identified during the recovery operation. The recovery module 132 determines the corrected rack number, the shelf number, and the slot position corresponding to the selected candidate laundry bag. At step 328, the recovery module 132 updates corrected rack-location information through the centralized database module 124. The user interaction module 128 provides the corrected rack-location information to the corresponding user through the user device for subsequent collection of the laundry bag. The corrected rack-location information identifies the corrected rack number, the shelf number, and the slot position associated with the selected candidate laundry bag.

[00135] According to an exemplary embodiment of the invention, FIG. 4 refers to a state transition diagram of the system 100 for tokenless laundry bag identity generation, localization, verification, and recovery. At step 402, the identified laundry bag remains available at the displayed rack location for collection by the corresponding user. The displayed rack location corresponds to the rack number, the shelf number, and the slot position determined by the AI-based rack localization module 126. The user interaction module 128

provides the rack number, the shelf number, the slot position, laundry-processing status, and collection availability status to the corresponding user through the user device. Retrieval of the laundry bag from the displayed rack location transitions the state transition diagram 400 from step 402 to step 404.

[00136] At step 404, the collection verification module 130 performs verification of the retrieved laundry bag. The collection verification module 130 compares current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity. The current visual and physical characteristics correspond to image appearance, colour distribution, shape, dimensions, texture characteristics, and weight.

[00137] In one embodiment herein, the collection verification module 130 determines the feature similarity score from comparison of the current visual characteristics with corresponding characteristics represented by the Laundry Identity Profile. The collection verification module 130 further determines the weight similarity score based on an originally recorded weight of the laundry bag and the current measured weight of the retrieved laundry bag. The collection verification module 130 generates the confidence score based on the feature similarity score and the weight similarity score according to respective weighting constants and evaluates the confidence score relative to the predefined threshold.

[00138] At step 406, satisfaction of the predefined threshold establishes successful verification of the retrieved laundry bag. The collection verification module 130 authorizes collection of the retrieved laundry bag based on the successful verification. Successful verification confirms correspondence between the retrieved laundry bag and the Laundry Identity Profile associated with the corresponding user identity. At step 408, completion of the authorized collection results in updating of a collection status associated with the laundry bag through the centralized database module 124. The collection status records completion of collection following successful verification of the retrieved laundry bag.

[00139] At step 410, failure of the confidence score associated with the retrieved laundry bag to satisfy the predefined threshold establishes the mismatch condition. The mismatch condition is also established upon detection of characteristics of the retrieved laundry bag that do not sufficiently correspond to the stored Laundry Identity Profile. The recovery

module 132 initiates recovery following detection of the mismatch condition. At step 412, the recovery module 132 initiates rack-wide scanning across the storage racks. The one or more overhead cameras acquire rack image data from respective sections of the storage racks containing the plurality of laundry bags positioned without the predefined sorting sequence. The recovery module 132 receives information associated with candidate laundry bags detected from the rack image data and performs recovery processing based on the stored Laundry Identity Profile.

[00140] At step 414, the recovery module 132 performs similarity matching between characteristics of the candidate laundry bags and the stored Laundry Identity Profile. The similarity matching evaluates correspondence between the visual and physical characteristics associated with each candidate laundry bag and the visual and physical characteristics represented by the Laundry Identity Profile. The recovery module 132 determines respective confidence scores associated with the candidate laundry bags.

[00141] At step 416, selection of the candidate laundry bag having the highest confidence score establishes re-identification of the laundry bag at a new rack location. The recovery module 132 determines the corrected rack number, the shelf number, and the slot position corresponding to the re-identified laundry bag. The corrected rack number, shelf number, and slot position establish corrected rack-location information associated with the laundry bag. At step 418, the recovery module 132 updates the corrected rack-location information through the centralized database module 124. The updated rack-location information replaces the previously determined rack-location information associated with the laundry bag. The user interaction module 128 provides the corrected rack-location information to the corresponding user through the user device.

[00142] At step 420, the corrected rack-location information identifies the new rack location for subsequent collection of the laundry bag. The corresponding user proceeds to the corrected rack number, the shelf number, and the slot position based on the corrected rack-location information provided through the user device. Retrieval of the laundry bag from the corrected rack location returns the verification flow to step 404 for verification of the retrieved laundry bag by the collection verification module 130.

[00143] At step 422, absence of the candidate laundry bag satisfying the recovery search results in a not-found state. The recovery flow proceeds to manual intervention following the not-found state. Following identification of corrected rack-location information through the manual intervention, the corrected rack-location information is updated through the centralized database module 124 and provided to the corresponding user through the user interaction module 128. In one embodiment herein, the recovery module 132 further initiates recovery in response to the missing-bag report received through the user device. The recovery module 132 performs rack-wide scanning, similarity matching, candidate confidence-score evaluation, and re-identification based on the stored Laundry Identity Profile for determination of corrected rack-location information.

[00144] According to an exemplary embodiment of the invention, FIGs. 5A and 5B refer to a flowchart 500 of a method for tokenless laundry bag identity generation, localization, verification, and recovery using the system 100. At step 502, the registration module 118 associates the laundry bag with the corresponding user identity based on identification information received through the user device. The identification information establishes the association between the laundry bag and the corresponding user identity for subsequent identity generation, localization, verification, and recovery operations.

[00145] At step 504, the feature acquisition module 120 acquires visual and physical characteristics of the laundry bag through the one or more cameras and the at least one weight sensor. The visual and physical characteristics are image appearance, colour distribution, shape, dimensions, texture characteristics, and weight. The acquired visual and physical characteristics provide input information for generation of the Laundry Identity Profile associated with the laundry bag.

[00146] At step 506, the artificial intelligence (AI) identity generation module 122 extract features from the visual and physical characteristics and converts the extracted features into numerical feature vectors. The artificial intelligence (AI) identity generation module 122 generates the Laundry Identity Profile from the numerical feature vectors. The centralized database module 124 stores the Laundry Identity Profile in association with the corresponding user identity, laundry-processing status, rack-location information, collection history, and verification and recovery records.

[00147] At step 508, the AI-based rack localization module 126 receives rack image data acquired by the one or more overhead cameras arranged to monitor storage racks containing the plurality of laundry bags positioned without the predefined sorting sequence. The rack image data represents respective sections of the storage racks and provides visual information associated with the laundry bags positioned across the storage racks. At step 510, the AI-based rack localization module 126 performs multi-object laundry bag detection on the rack image data and extracts visible characteristics of detected laundry bags. The AI-based rack localization module 126 compares the visible characteristics of the detected laundry bags with stored Laundry Identity Profiles using similarity matching and identifies the laundry bag based on the highest similarity match.

[00148] At step 512, the AI-based rack localization module 126 determines the rack number, the shelf number, and the slot position corresponding to the identified laundry bag. The rack number, the shelf number, and the slot position establish location information associated with the identified laundry bag. The user interaction module 128 provides the location information associated with the identified laundry bag to the corresponding user through the user device.

[00149] At step 514, the collection verification module 130 verifies the retrieved laundry bag based on comparison of current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity. The collection verification module 130 determines the feature similarity score and the weight similarity score and generates the confidence score based on the feature similarity score and the weight similarity score. The collection verification module 130 authorizes collection of the retrieved laundry bag when the confidence score satisfies the predefined threshold.

[00150] At step 516, the recovery module 132 initiates rack-wide scanning of the storage racks upon detection of the at least one condition selected from absence of the laundry bag from the determined rack location and failure of the confidence score to satisfy the predefined threshold. The one or more overhead cameras acquire rack image data during the rack-wide scanning. The recovery module 132 compares candidate laundry bags detected during the rack-wide scanning with the stored Laundry Identity Profile using similarity

matching. The recovery module 132 determines respective confidence scores associated with the candidate laundry bags.

[00151] At step 518, the recovery module 132 selects the candidate laundry bag having the highest confidence score and determines the corrected rack number, the shelf number, and the slot position of the selected candidate laundry bag. The recovery module 132 updates corrected rack-location information through the centralized database module 124. The user interaction module 128 provides the corrected rack-location information to the corresponding user through the user device for subsequent collection of the laundry bag.

[00152] In accordance with the present disclosure, the system 100 performs tokenless laundry bag identity generation based on visual and physical characteristics acquired through the one or more cameras and the at least one weight sensor. The feature acquisition module 120 converts the acquired characteristics into machine-readable feature representations, and the artificial intelligence (AI) identity generation module 122 processes the machine-readable feature representations to generate numerical feature vectors forming the Laundry Identity Profile associated with the corresponding user identity. The Laundry Identity Profile provides a stored feature-based representation of the physical laundry bag for subsequent rack localization, collection verification, re-identification, and recovery.

[00153] The system 100 performs autonomous localization of laundry bags positioned on storage racks without the predefined sorting sequence based on rack image data continuously acquired through the one or more overhead cameras. The AI-based rack localization module 126 performs multi-object laundry bag detection, extracts visible characteristics of the detected laundry bags, compares the visible characteristics with stored Laundry Identity Profiles using similarity matching, identifies the laundry bag based on the highest similarity match, and determines the corresponding rack number, shelf number, and slot position. The determined rack-location information is stored through the centralized database module 124 and provided through the user interaction module 128.

[00154] The system 100 performs collection verification by comparing current visual and physical characteristics of the retrieved laundry bag with the stored Laundry Identity Profile associated with the corresponding user identity. The collection verification module 130 determines the feature similarity score and the weight similarity score, generates the

confidence score according to respective weighting constants, and authorizes collection when the confidence score satisfies the predefined threshold. Failure of the confidence score to satisfy the predefined threshold establishes the mismatch condition for initiating rack-wide recovery.

[00155] Upon detection of the mismatch condition or absence of the laundry bag from the determined rack location, the recovery module 132 initiates rack-wide scanning of the storage racks, compares candidate laundry bags with the stored Laundry Identity Profile, determines respective confidence scores, selects the candidate laundry bag having the highest confidence score, and determines corrected rack-location information comprising the corrected rack number, shelf number, and slot position. The corrected rack-location information is updated through the centralized database module 124 and provided through the user interaction module 128. The cooperative operation of the physical image acquisition, weight acquisition, feature-based identity generation, rack localization, collection verification, and recovery operations enables the system 100 to identify, localize, verify, and recover laundry bags independently of RFID tags, barcode labels, QR codes, and other physical identifiers.

[00156] It will readily be apparent that numerous modifications and alterations can be made to the processes described in the foregoing examples without departing from the principles underlying the invention, and all such modifications and alterations are intended to be embraced by this application.

5. CLAIMS:

I/We Claim:

1. A system (100), comprising:

a computing device (102) having at least one processor (104) and a memory (106) for storing one or more instructions executable by the at least one processor (104),

wherein the computing device (102) is communicatively coupled to a server (108) via a network (110),

wherein the computing device (102) is communicatively coupled to one or more cameras and at least one weight sensor arranged to acquire visual and physical characteristics of a laundry bag, and

wherein the at least one processor (104) is configured to execute plurality of modules (116) for identifying, localizing, verifying, and recovering the laundry bag based on the acquired visual and physical characteristics of the laundry bag,

wherein the plurality of modules (116) comprises:

a registration module (118) configured to associate the laundry bag with a corresponding user identity based on at least one of a registration number, an identity card, and authentication through a user device;

a feature acquisition module (120) configured to receive the visual and physical characteristics of the laundry bag acquired through the one or more cameras and the at least one weight sensor, wherein the visual and physical characteristics comprise image appearance, colour distribution, shape, dimensions, texture characteristics, and weight;

an artificial intelligence (AI) identity generation module (122) configured to extract features from the visual and physical characteristics, convert the extracted features into numerical feature vectors, generate a Laundry Identity Profile associated with the corresponding user identity, assign respective weighting factors to the extracted features, and combine similarity values corresponding to the extracted features according to the respective weighting factors for determining a feature similarity score;

a centralized database module (124) configured to store the Laundry Identity Profile in association with the corresponding user identity, laundry-processing status, rack-location information, collection history, and verification and recovery records;

an AI-based rack localization module (126) operatively connected to the centralized database module (124), wherein the AI-based rack localization module (126) is configured to:

receive rack image data acquired by one or more overhead cameras arranged to monitor storage racks containing a plurality of laundry bags positioned without a predefined sorting sequence;

perform multi-object laundry bag detection on the rack image data;

extract visible characteristics of detected laundry bags;

compare the visible characteristics of the detected laundry bags with stored Laundry Identity Profiles using similarity matching and identify the laundry bag based on the similarity matching; and

determine a rack number, a shelf number, and a slot position corresponding to the identified laundry bag;

a user interaction module (128) configured to receive the rack number, the shelf number, and the slot position determined by the AI-based rack localization module (126), and provide laundry-processing status, collection availability status, and location information associated with the laundry bag to the corresponding user through the user device;

a collection verification module (130) configured to compare current visual and physical characteristics of a retrieved laundry bag with the Laundry Identity Profile associated with the corresponding user identity, determine the feature similarity score and a weight similarity score, generate a confidence score based on the feature similarity score and the weight similarity score, and authorize collection of the retrieved laundry bag when the confidence score satisfies a predefined threshold; and

a recovery module (132) configured to:

initiate rack-wide scanning of the storage racks in response to at least one of absence of the laundry bag from a determined rack location and failure of the confidence score associated with the retrieved laundry bag to satisfy the predefined threshold;

compare candidate laundry bags detected during the rack-wide scanning with the stored Laundry Identity Profile;

determine respective confidence scores associated with the candidate laundry bags;

select a candidate laundry bag having a highest confidence score;

determine a corrected rack number, shelf number, and slot position of the selected candidate laundry bag; and

update corrected rack-location information through the centralized database module (124) and provide the corrected rack-location information to the corresponding user through the user interaction module (128) for subsequent collection of the laundry bag,

wherein identification, localization, verification, and recovery of the laundry bag are performed independently of an RFID tag, a barcode label, a QR code, and other physical identifiers associated with the laundry bag.

2. The system (100) as claimed in claim 1, wherein the artificial intelligence (AI) identity generation module (122) is configured to assign respective weighting factors to the extracted features and combine similarity values of the extracted features according to the respective weighting factors for determining the feature similarity score.

3. The system (100) as claimed in claim 1, wherein the one or more overhead cameras are configured to continuously acquire rack image data from respective sections of the storage racks, and wherein the AI-based rack localization module (126) is configured to identify the laundry bag based on a highest similarity match between the visible characteristics of the detected laundry bags and the stored Laundry Identity Profiles.

4. The system (100) as claimed in claim 1, wherein the user interaction module (128) is configured to provide the corresponding user with laundry submission confirmation,

laundry-processing status, laundry-completion notification, the rack number, the shelf number, the slot position, and collection availability status associated with the laundry bag.

5. The system (100) as claimed in claim 1, wherein the collection verification module (130) is configured to determine the weight similarity score based on a difference between an originally recorded weight of the laundry bag and a current measured weight of the retrieved laundry bag relative to the originally recorded weight.

6. The system (100) as claimed in claim 5, wherein the collection verification module (130) is configured to generate the confidence score by combining the feature similarity score and the weight similarity score according to respective weighting constants.

7. The system (100) as claimed in claim 1, wherein the collection verification module (130) is configured to update a collection status associated with the laundry bag through the centralized database module (124) upon authorization of collection based on satisfaction of the predefined threshold.

8. The system (100) as claimed in claim 1, wherein the recovery module (132) is configured to initiate recovery automatically upon detection of at least one condition selected from absence of the laundry bag from the determined rack location and a mismatch between characteristics of the retrieved laundry bag and the stored Laundry Identity Profile, and to initiate recovery in response to a missing-bag report received through the user device.

9. The system (100) as claimed in claim 1, wherein the centralized database module (124) is configured to synchronize the Laundry Identity Profile, the laundry-processing status, the rack-location information, the collection history, and the verification and recovery records with the AI-based rack localization module (126) and the user interaction module (128).

10. A method for tokenless laundry bag identity generation, localization, verification and recovery using a system (100), comprising:

associating, by a registration module (118), a laundry bag with a corresponding user identity based on identification information received through a user device;

acquiring, by a feature acquisition module (120), visual and physical characteristics of the laundry bag through one or more cameras and at least one weight sensor;

generating, by an artificial intelligence (AI) identity generation module (122), a Laundry Identity Profile from numerical feature vectors derived from the visual and physical characteristics of the laundry bag, and storing the Laundry Identity Profile through a centralized database module (124);

determining, by the artificial intelligence (AI) identity generation module (122), a feature similarity score by assigning respective weighting factors to extracted features and combining similarity values corresponding to the extracted features according to the respective weighting factors;

receiving, by an AI-based rack localization module (126), rack image data acquired by one or more overhead cameras arranged to monitor storage racks containing a plurality of laundry bags positioned without a predefined sorting sequence;

detecting, by the AI-based rack localization module (126), laundry bags from the rack image data, comparing visible characteristics of the detected laundry bags with stored Laundry Identity Profiles using similarity matching, and identifying the laundry bag based on a highest similarity match;

determining, by the AI-based rack localization module (126), a rack number, a shelf number, and a slot position corresponding to the identified laundry bag, and providing, by a user interaction module (128), location information associated with the identified laundry bag to the corresponding user through the user device;

verifying, by a collection verification module (130), a retrieved laundry bag based on comparison of current visual and physical characteristics of the retrieved laundry bag with the Laundry Identity Profile, generating a confidence score based on the feature similarity score and a weight similarity score, and authorizing collection of the retrieved laundry bag when the confidence score satisfies a predefined threshold;

initiating, by a recovery module (132), rack-wide scanning of the storage racks upon detection of at least one condition selected from absence of the laundry bag from a determined rack location and failure of the confidence score to satisfy the predefined threshold, comparing candidate laundry bags detected during the rack-wide scanning with

the stored Laundry Identity Profile, and determining respective confidence scores associated with the candidate laundry bags; and

selecting, by the recovery module (132), a candidate laundry bag having a highest confidence score, determining a corrected rack number, the shelf number, and the slot position of the selected candidate laundry bag, updating corrected rack-location information through the centralized database module (124), and providing the corrected rack-location information to the corresponding user through the user interaction module (128),

wherein the identity generation, localization, verification, and recovery of the laundry bag are performed based on the visual and physical characteristics of the laundry bag independently of an RFID tag, a barcode label, a QR code, and other physical identifiers associated with the laundry bag.

6. DATE AND SIGNATURE:

Dated this 29th day of August, 2026



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7. ABSTRACT:

Title: A System and Method for Tokenless Laundry Bag Identity Generation, Localization, Verification and Recovery

The present disclosure proposes a system (100) and method for tokenless laundry bag identity generation, autonomous rack localization, collection verification, and misplaced laundry bag recovery in large-scale laundry environments. The system (100) comprises a computing device (102), at least one processor (104), a memory (106), and plurality of modules (116) executed by the at least one processor (104). The system (100) acquires visual and physical characteristics of a laundry bag, generates a Laundry Identity Profile from numerical feature vectors, and localizes the laundry bag on storage racks through rack image data and similarity matching. The system (100) verifies a retrieved laundry bag using a feature similarity score, a weight similarity score, and a confidence score. Upon verification failure or absence from a determined rack location, the system (100) performs rack-wide scanning, identifies a matching laundry bag, and updates corrected rack-location information independently of physical identifiers.

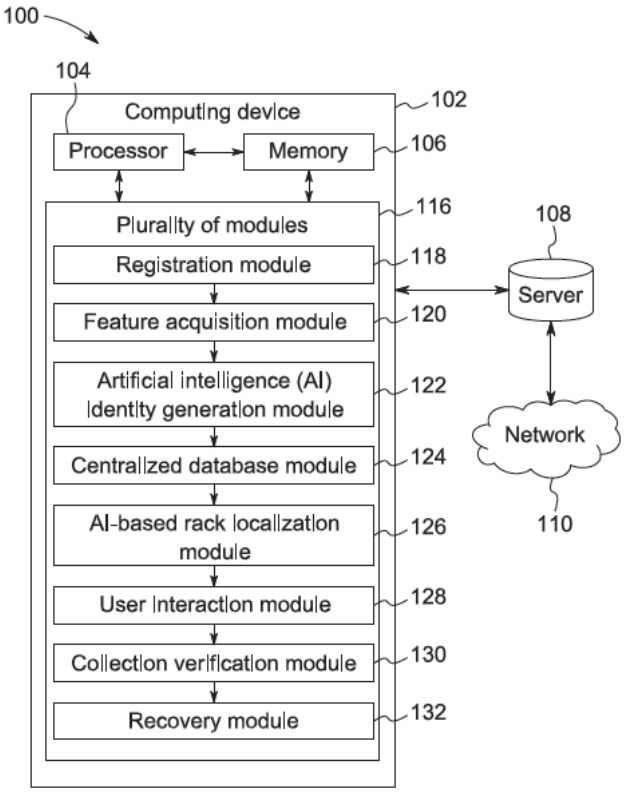


FIG. 1

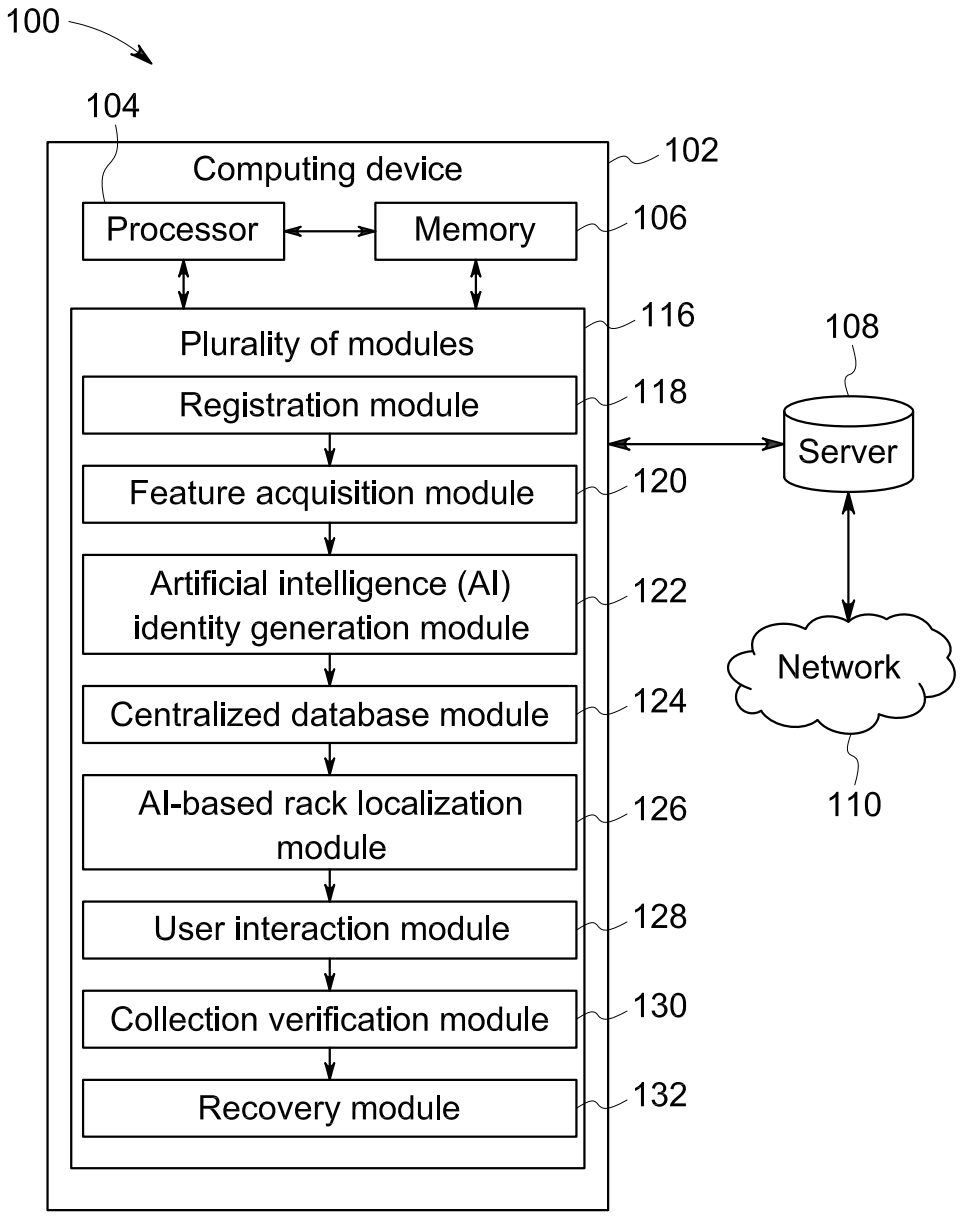


FIG. 1

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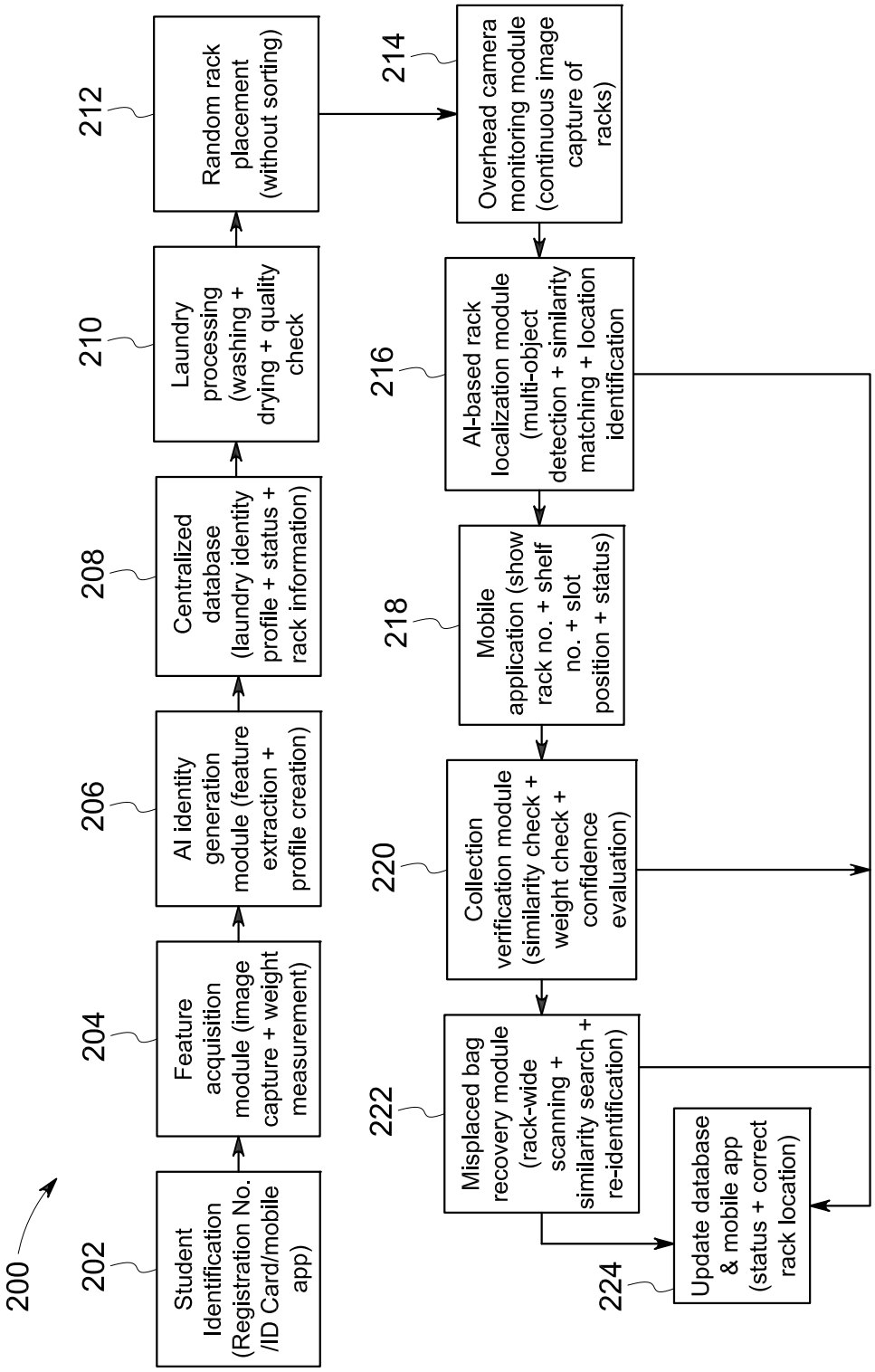


FIG. 2

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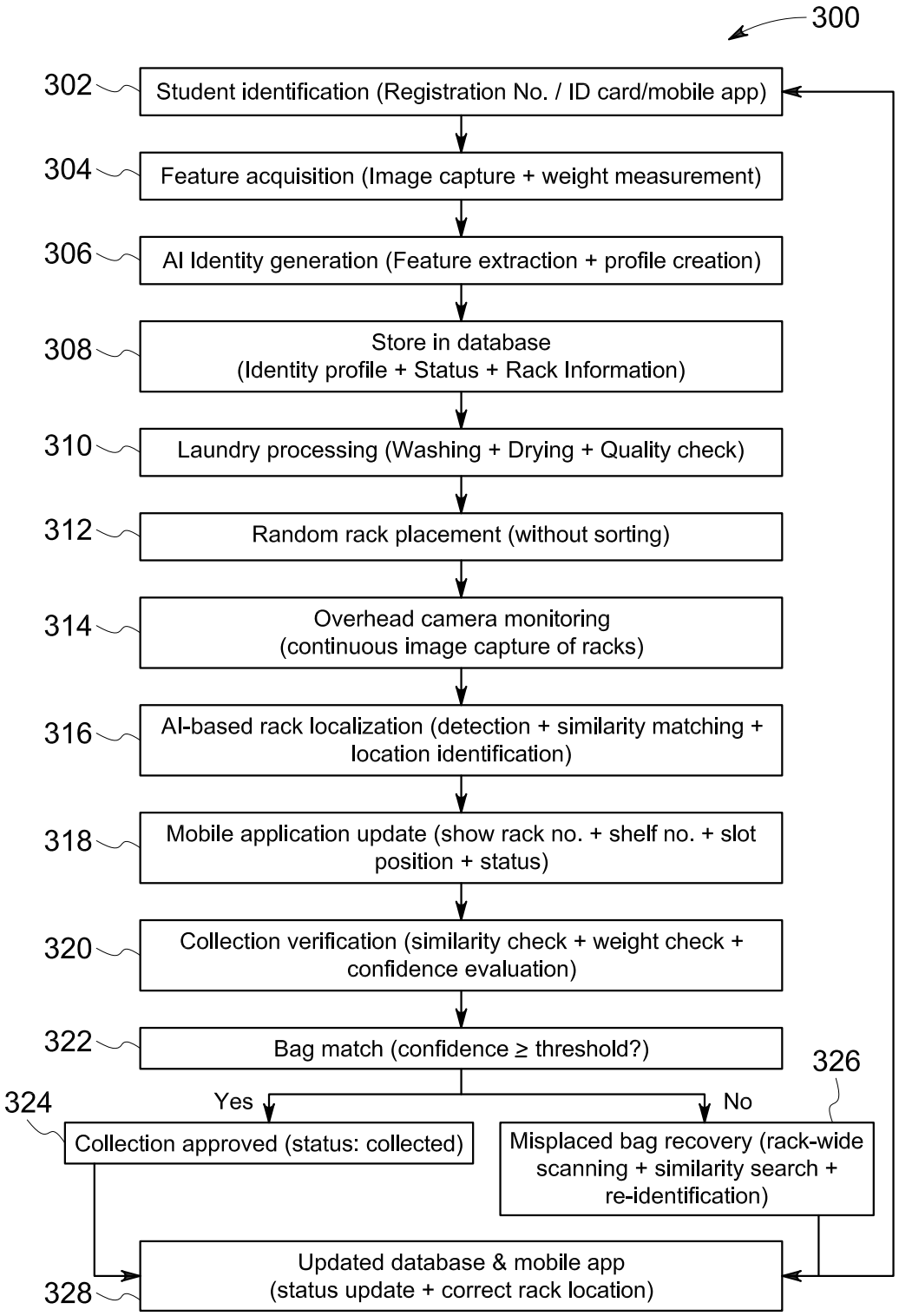


FIG. 3

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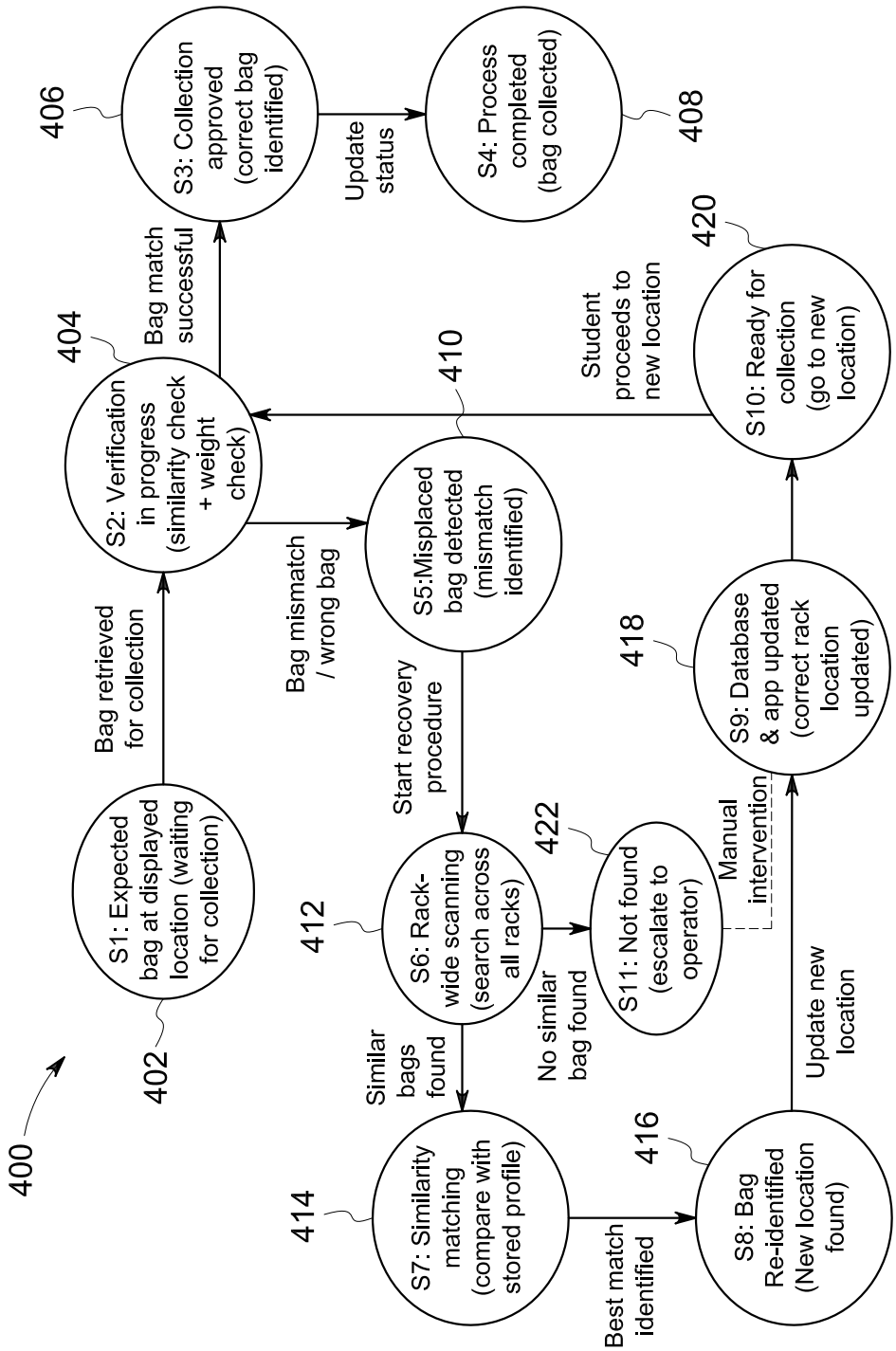


FIG. 4

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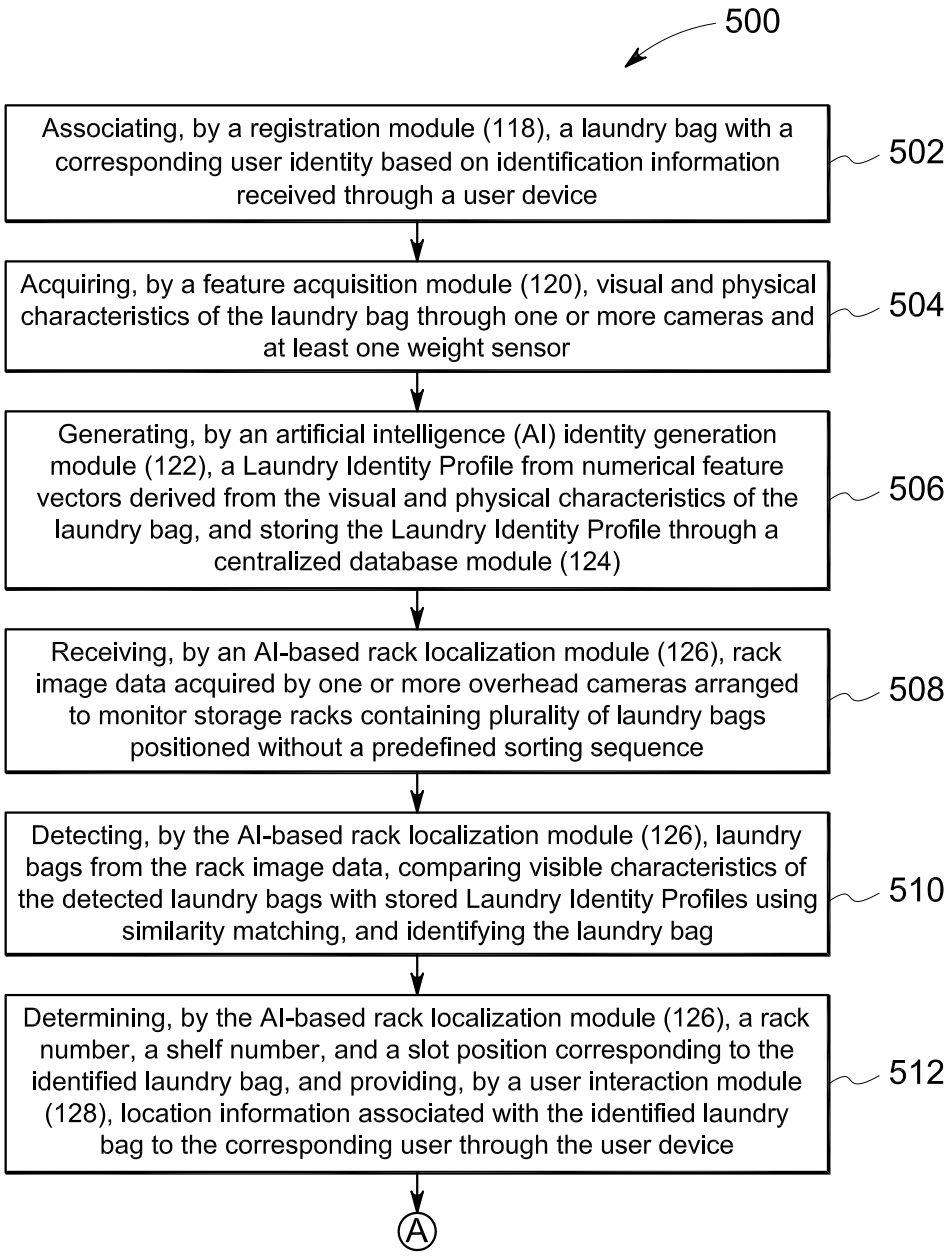


FIG. 5A

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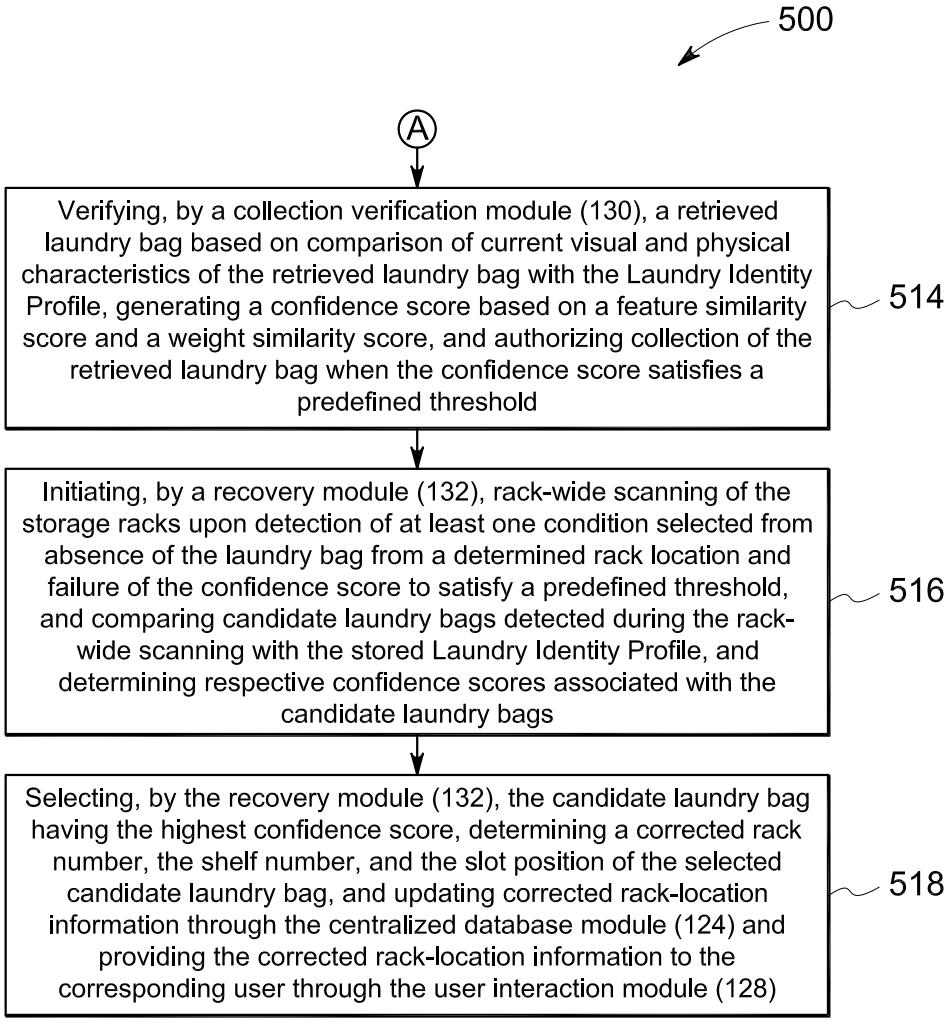


FIG. 5B

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